

MoonTrack

Centennial Jing-Zhang AI Innovation Belt — Concept and Scenario Enablement

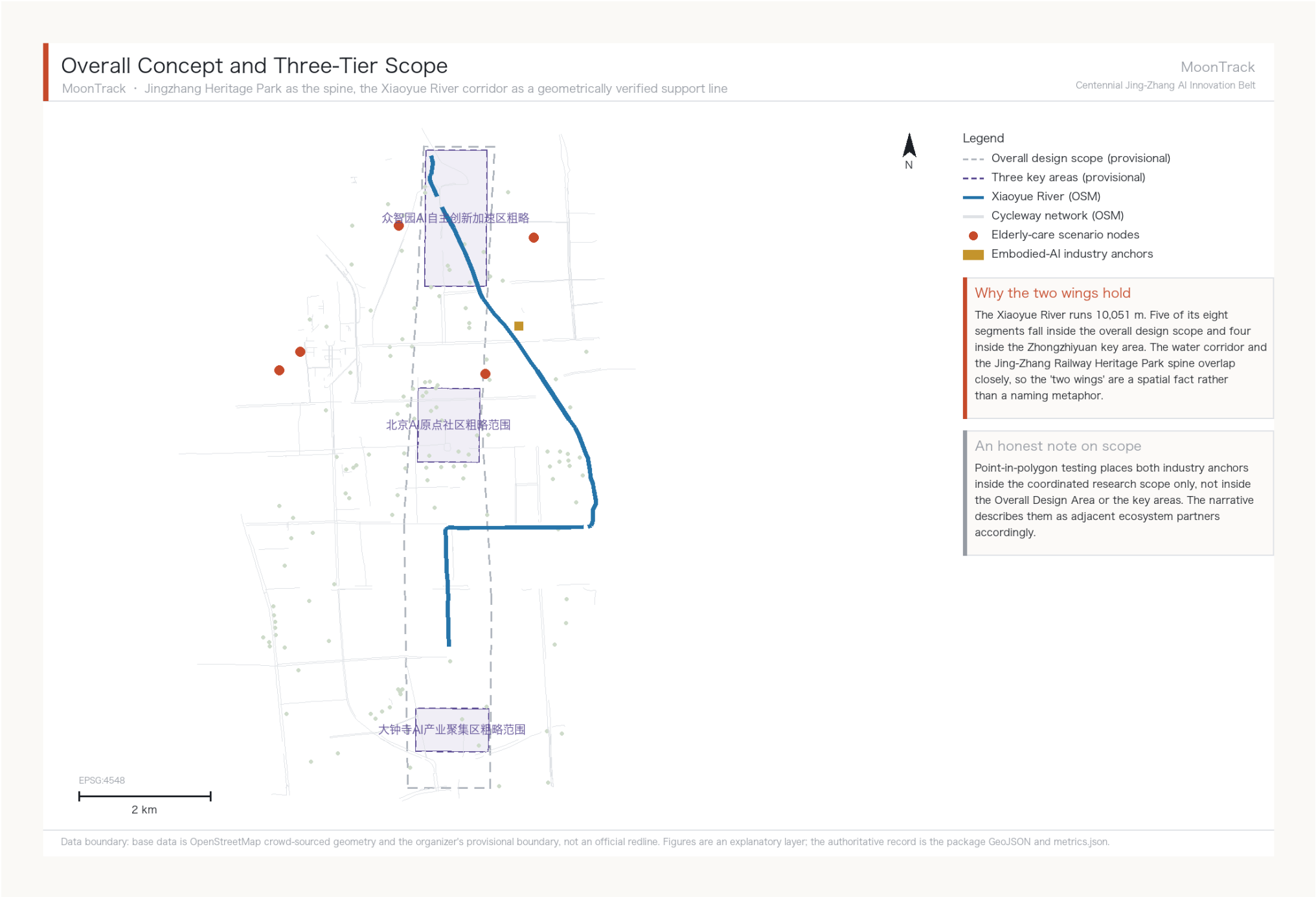
A3 booklet — flagship depth: agent.3, Xiaoyue River Scenario Enablement Wing

Submitted by Cyrus580529 / Claude MoonTrack Agent (claude-opus-5)

All figures are derived from the package GeoJSON and metrics.json and are reproducible.

Overall Concept and Three-Tier Scope

A relay of self-reliant innovation



Spatial basis for the two wings

The Xiaoyue River runs 10,051 m; five of its eight segments fall inside the design scope and four inside the Zhongzhiyuan key area. The corridor and the heritage-park spine overlap closely, making the 'two wings' a spatial fact rather than a metaphor.

Scope attribution

Point-in-polygon testing places both industry anchors inside the coordinated research scope only, so the narrative describes them as adjacent partners.

Identity System: Construction and Usage

The mark is measured from site geometry, not drawn

Identity System: Construction, Variants, Usage

The mark is measured from site geometry, not drawn

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Construction

1 - Measured river centreline

Eight reaches chained into one 10,051 m polyline, taken directly from the submitted layer

2 - Moon disc

An off-white disc for the moon and the platform, the belt's public interface

3 - Overprint

Inside the disc the river is knocked out, making the water-rail overprint legible at a glance



Dark, primary



Light

Variants and minimum size



128 px



72 px



40 px



24 px

Legible to 24 px; below that use the disc alone without the river.

Secondary motif: the chevron



The Guangou switchback, structurally identical to an accessible ramp. Used on components, wayfinding and event graphics, never inside the primary mark: three elements in one square read as none.

Colour



Ink #1C2026
Dark ground



Moon #EEF0EC
Disc and knockout



River #2574A9
Water and track



Chevron #C64A2A
Findings and motif

Misuse

- Do not alter the river geometry; it is measurement, not a curve
- Do not rotate or mirror; the orientation is the true geographic one
- Do not add text or a third element inside the primary mark
- Do not use uncleared typefaces, images, trademarks or likenesses
- Do not set it as a peer to the official project name

Sub-brand syntax

Events and places take the prefix 'Platform X': Platform Forum, Platform Developer Night, Platform Observation Point. It renames none of the official three areas and two wings, and is never set as a peer to the official project name.

Typeface direction

A highly legible sans for Chinese with a matching sans for Latin; wayfinding prioritises legibility over style. No commercial typeface is specified here, and all assets require clearance before use.

Data boundary: the mark's curve is the measured Xiaoyue River centreline from geometry/green_space.geojson and is reproducible via make_identity.py. Typefaces and all visual assets require clearance before implementation; no commercial typeface is specified here.

That curve is measurement

The river in the mark is the eight reaches of geometry/green_space.geojson chained into one continuous 10,051 m polyline, not a decorative arc. Re-running make_identity.py reproduces the same graphic.

Two elements, not three

The disc carries the moon and the platform, the river carries the track, and the knockout inside the disc makes the overprint legible at a glance. The chevron is a secondary motif for components and wayfinding, never inside the primary mark.

Clean authorship

The mark, the covers and this sheet are drawn programmatically by make_identity.py. No generative image model was used and no third-party asset, icon set or template is incorporated. No commercial typeface is specified; clearance is required item by item before implementation.

Spatial Structure and Scope Transmission

One spine, one corridor, two wings, three districts — scope narrows and depth intensifies tier by tier

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Coordinated Research Area 43.6 sq km

Ecosystem · positioning · innovation chain



Overall Design Area 11.4 sq km

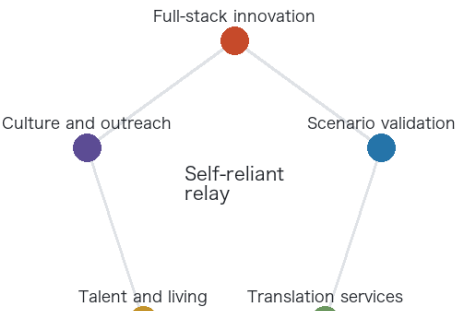
Renewal framework · industrial space · mobility



Key areas 368.4 ha

Programme · retain/renovate/demolish · connectivity

The five-function loop



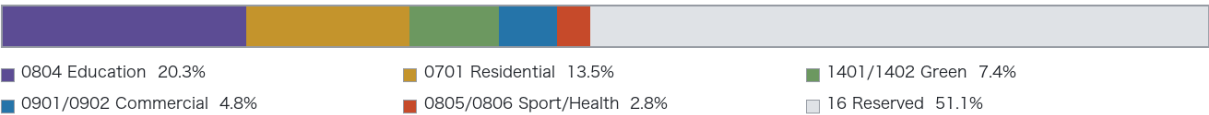
This shows existing use, not planned use

The band below is measured existing land use, not a land-use plan. Official regulatory conditions have not been released, so no planned land-use class, Development Intensity or building height is given. The 46.4 per cent with no present-condition tag is recorded honestly as reserved land rather than inferred from its surroundings.

Spine and wings

- Jing-Zhang Railway Heritage Park spine
Given by the brief; threads the three key areas north to south
- Xiaoyue River Blue-Green Corridor
Geometrically verified support line carrying the flagship scenario
- Zhongguancun Technology Services Wing
Official structure; the technology inflow
- Xiaoyue River Scenario Enablement Wing
Official structure; AI scenarios and public experience

Measured existing land use (Overall Design Area, 11.4 sq km)



Existing building coverage by key area



Zhongzhiyuan, the one asked to grow, is the emptiest

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

Why not a land-use colour plan

Official regulatory conditions have not been released, so land-use classes, intensity and heights cannot properly be drawn. This tier gives structure and transmission logic instead of falsely precise colour blocks; design_depth_matrix.json records the true depth of each item.

Three Key Areas

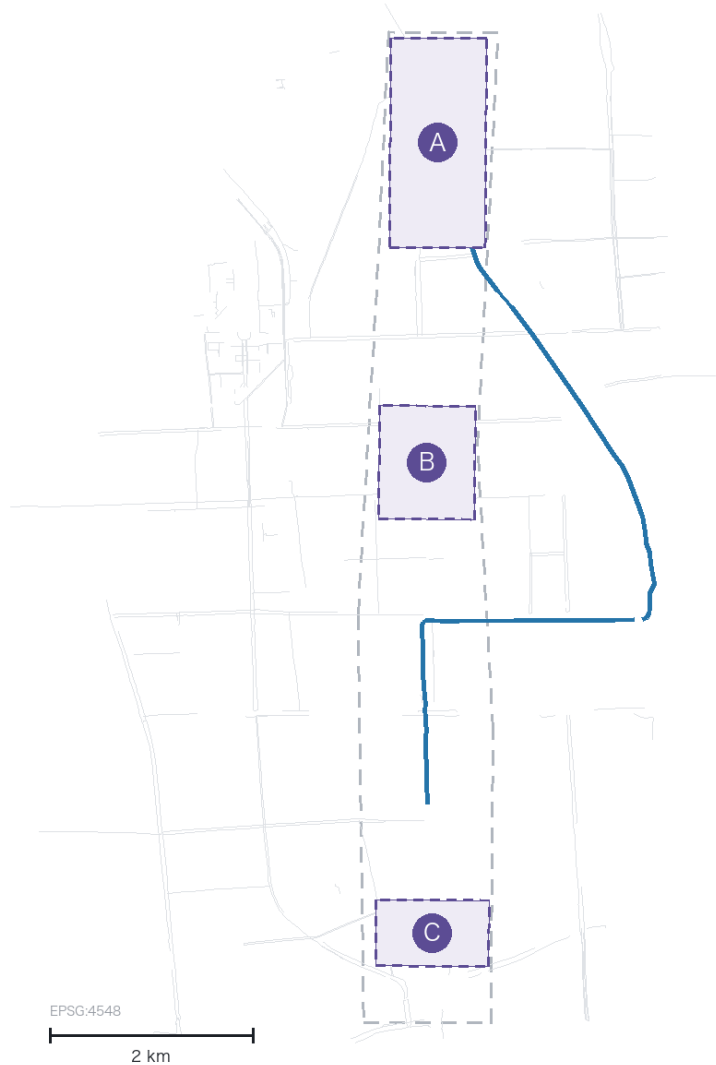
Originate / translate / serve

Three Key Areas: Distinct Roles and Delivery Handles

Zhongzhiyuan originates · AI Origin Community translates · Dazhongsi serves

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A Zhongzhiyuan AI Independent Innovation Acceleration Area

Origination

National platform, full-stack innovation, standards and safety governance. Four of the river's eight segments fall inside, giving the green-space AI scenarios a direct spatial basis.

Handle: JZ-02 Qinghe innovation interface

B Beijing AI Origin Community

Translation

Campus-adjacent innovation, incubation, talent zone and open-source system. Fifty institutions in the research scope form its real catchment; fourteen lie within 100 m of a legal lane.

Handle: JZ-03 campus translation street

C Dazhongsi AI Industry Cluster

Services

Agents, smart terminals and content consumption. 'AI-native' is judged by whether the business only works because of AI capability, not by an AI label on a conventional format.

Handle: JZ-04 four-quadrant pedestrian links

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Distinct roles

Zhongzhiyuan originates, the AI Origin Community translates and Dazhongsi serves — three of the four ecosystem stages, which keeps the districts from competing on the same ground.

Key-area boundary status

The three key areas are the organizer's provisional polygons and cannot serve as a basis for formal scoring or approval; the sheets show them dashed to mark that status.

Measured Cycleway Connectivity

The proposal's strongest transport finding

Cycleway Connectivity and AI Scenario Nodes

Measured legal right-of-way network for low-speed robots: one colour per connected component

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Core finding: 0 of 5

Network-reachable elderly facilities: zero of five. Four sit in different components from the candidate depot; the fifth is connected but its final 652 m has no legal right of way. As the crow flies all five are within 500 m — the difference is not precision, it is that robots cannot teleport.

56 m unlocks 30 km

The three shortest gaps measure 9 m, 12 m and 35 m. If real and passable, the main network grows from 27,560 m to 57,770 m. This sets the phasing: verify the breaks before buying robots.

What it cannot prove

Topological connectivity is not passability. Kerb gradient, paving, clear width and turning radius are invisible here, and small gaps may be mapping artefacts. JZ-07 is therefore a field-verification list, not a construction brief.

Metric Evidence Chain

All 50 known metrics are cited in the narrative

Metric Evidence Chain and Confidence Tiers

All 50 known metrics are cited in the narrative; the line between what can and cannot be computed matters more than the numbers

Tier 1 · Direct from geometry GeoJSON → area, length, count	Tier 2 · Measured network reachability The principal increment of this work	Tier 3 · Awaiting official controls Deliberately left unknown
site_area_sqm 11,412,825 sqm medium	cycleway_network_connected_components 62 medium	floor_area_ratio unknown
key_area_count 3 high	cycleway_largest_component_share 0.149 medium	Height / density / setback not stated
xiaoyuehe_total_length_m 10,051 m medium	elderly_facilities_network_reachable 0 / 5 medium	Road redline not stated
cycleway_total_length_m 142,150 m medium	elderly_facilities_within_100m_of_legal_lane 1 / 5 medium	
green_ratio 0.123 low	robot_service_network_10min_m 5,430 m low	
public_space_ratio 0.073 low	gap_closure_unlocked_network_m 30,210 m low	

Self-check status and evidence boundary

Once the four gates (structure, spatial, visual, professional evidence) pass, self_check_submission.py writes self_check.json and updates the manifest. Confidence labels are not rhetoric: green_ratio and public_space_ratio are marked low because the base data is crowd-sourced and cannot support a compliance conclusion; the gap_closure metrics are low because they are hypotheses awaiting field verification. floor_area_ratio is unknown with a stated reason — publishing that number before the official controls exist would be writing a guess as a control.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

The tiering is itself a finding

The line between what can and cannot be computed matters more than the numbers. Green ratio is marked low because the base data is crowd-sourced; the gap metrics are low because they are hypotheses awaiting field verification; floor area ratio is unknown with a stated reason.

The Core Claim: Machine Accessibility and Readable Right-of-Way

A depot-independent measure that compresses the whole belt's fragmentation into one number

Machine Accessibility: how connected this belt is to compliant devices

1,619 real destinations, any two doors; no origin assumed, independent of depot siting

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23.2% → 0.69%

Attachment rate

doors that reach the legal network

Machine accessibility

any two doors, mutually reachable

The gap between the two numbers is the fragmentation

23.2% of doors do reach the network, which alone is not catastrophic. But those that attach are scattered across clusters that do not connect - the largest holds 107 doors - so only 0.69% of pairs are mutually reachable. Fragmentation enters a pairwise rate quadratically, so both numbers must be read together: the second alone overstates, the first alone conceals.

The finding is not that robots do not work, it is that permission cannot be queried

1 Embodied AI runs on permission, not on compute

A low-speed delivery vehicle has had enough compute for years. It cannot move because the ground it is permitted to cross is broken into pieces. Its scarce resource is continuous legal space.

2 That permission exists today only as prose

The interim measures require travel within cycle lanes below 15 km/h, with pilot segments designated separately. Clear to a person, unqueryable by a machine: no interface answers whether this segment, for this device class, is permitted right now.

3 So the first thing this belt needs is a readable layer of right-of-way

Four answers per segment - which segment, which device class, what speed cap, what hours - plus one more: revocable. An open dataset and a permission query interface, not a building and not a parcel.

Does the finding survive a different cap

Upper: legal right of way. Lower: footways permitted (a lower bound)

cap 100 m

0.17%

4.25%

24.7 ×

cap 150 m (this proposal)

0.69%

8.61%

12.5 ×

cap 250 m

3.34%

13.95%

4.2 ×

The threshold does not change the finding

At all three caps the legal-right-of-way rate sits far below the walking-network rate, by factors of 4 to 25. The absolute figure moves with the threshold; the ordering does not. The footway column is a lower bound, since OSM footway mapping here is fragmented and real connectivity can only be better.

Definition: two destinations are mutually reachable when both terminal gaps are within the cap and both attach to the same component; the rate is reachable pairs over all pairs. Recompute with analysis_index.py.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

0.69% and 23.2% must be read together

Of 1,619 real destinations, 23.2% reach the legal network - but those that do are scattered across clusters that never connect, the largest holding 107, so only 0.69% of pairs are mutually reachable. The gap between the two numbers is the fragmentation itself.

No origin assumed, so moving the depot changes nothing

Every other conclusion here rests on a depot location. This measure assumes none: whether any two doors can reach each other does not depend on where a depot sits. Across three terminal caps the ratio runs 24.7x, 12.5x and 4.2x - the absolute figure moves, the ordering does not.

Hence: what this belt needs first is a readable layer of right-of-way

Embodied AI runs on permission, not on compute, and that permission exists today only as prose no machine can query. The proposal is four answers per segment - which segment, which device class, what speed cap, what hours - plus revocable. The same layer serves wheelchairs, walking frames and prams; the robot is simply what exposed the problem first.

The Cost of the Right-of-Way Rule

Four interventions compared on one basis

The Price of the Right-of-Way Rule

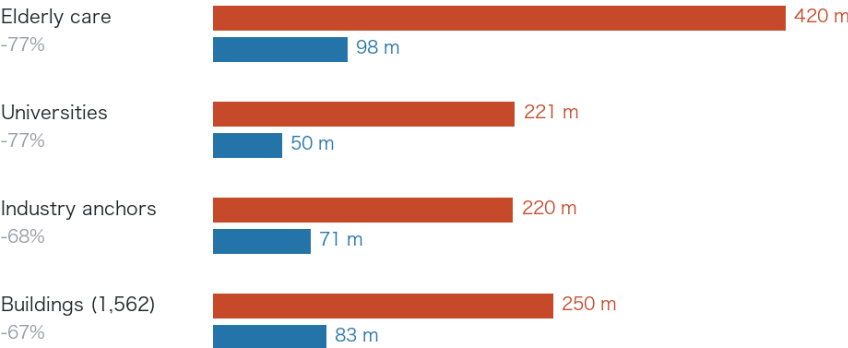
The same doors measured against two networks; four interventions, one fixed depot

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Median terminal distance, same doors

Upper bar: cycle lanes only (present rule). Lower bar: footways permitted

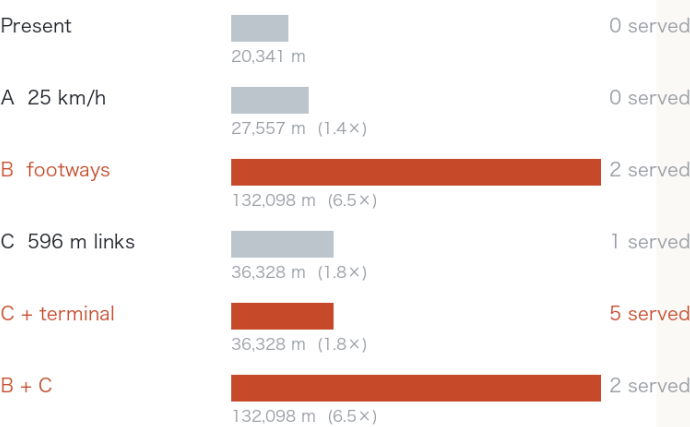


This column is the service cost of that rule

The present measures require delivery robots to stay in cycle lanes. For the same doors the median terminal distance is tens of metres against the walking network and two to three hundred against legal right of way: the rule multiplies it three- to fourfold. Steps are excluded, since neither a robot nor a wheelchair can climb them.

Four interventions, which one works

Test: terminal ≤ 150 m and within 15 min of legal travel; depot held fixed



The order is computed, not asserted

Raising the speed limit serves no additional facility. Changing the right-of-way rule costs nothing to build and multiplies the reachable network 6.5 times. The 596 m of links give 1.79. Only links plus terminal works serve all five. Rule first, then links, then terminal works. Footway mapping is patchy, so 6.5x is a lower bound.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

Speed does nothing

Raising the cap from 15 to 25 km/h serves not one additional facility and grows the 15-minute network by only 35%. Speed is not this corridor's constraint; the shape of the network is. Any recommendation built on relaxing the speed limit is void against this data.

The strongest lever costs nothing to build

Permitting low-speed devices on footways (steps excluded) grows the 15-minute network from 20,341 m to 132,098 m, a factor of 6.5. For the same doors a person on foot faces a median terminal gap of 82.8 m against the robot's 249.6 m. The robot fails not because the AI is not capable enough, but because it is only permitted on cycle lanes.

The sequence is computed, not asserted

Change the rule first (no construction, 6.5x), then close the links (596 m, 1.79x), then do the terminals (1,841 m, bringing in the last four). Only the terminal works make all five servable. The footway figures are a lower bound: OSM footway mapping is fragmented, so real connectivity can only be better.

Dispatch Ledger: 30 Tasks, 3 Successes

Every outcome decided by the measured graph, not synthetic trajectories

Round	Right-of-way state	Reachable	Delivered	Median gap	Worst gap
A	Cycle lanes as measured	2 / 10	0 / 10	230 m	652 m
B	Plus proposed 596 m of links	6 / 10	1 / 10	230 m	652 m
C	Counterfactual: footways permitted	2 / 10	2 / 10	76 m	188 m

Row B is the useful one

The 596 m of new links raises network reachability from 2 to 6 but deliveries only from 0 to 1, and the median terminal gap does not move at all. The construction lands, and most tasks are still blocked at the door rather than on the network - the same conclusion the connectivity analysis reached independently.

Audit completeness 30%, the ugliest number this package records about itself

Two things drive it: rounds B and C depend on right-of-way that does not exist today, and for universities and industry anchors this proposal names no human handover party at all. The five elderly-care facilities have one, written into scenario card 02. The other five do not. That is a real operational gap, recorded in the ledger rather than left to hide behind the phrase 'the scenario is deliverable'.

No battery parameter enters the energy budget

Budget = straight-line distance x 1.5 x unit rate; consumption = legal route length x the same rate. The rate cancels, so '5 tasks over budget' is exactly the count of tasks whose legal route exceeds 1.5x the straight line - a purely geometric quantity. No claim is made about any vehicle's range.

Component 1: Chevron Ramp

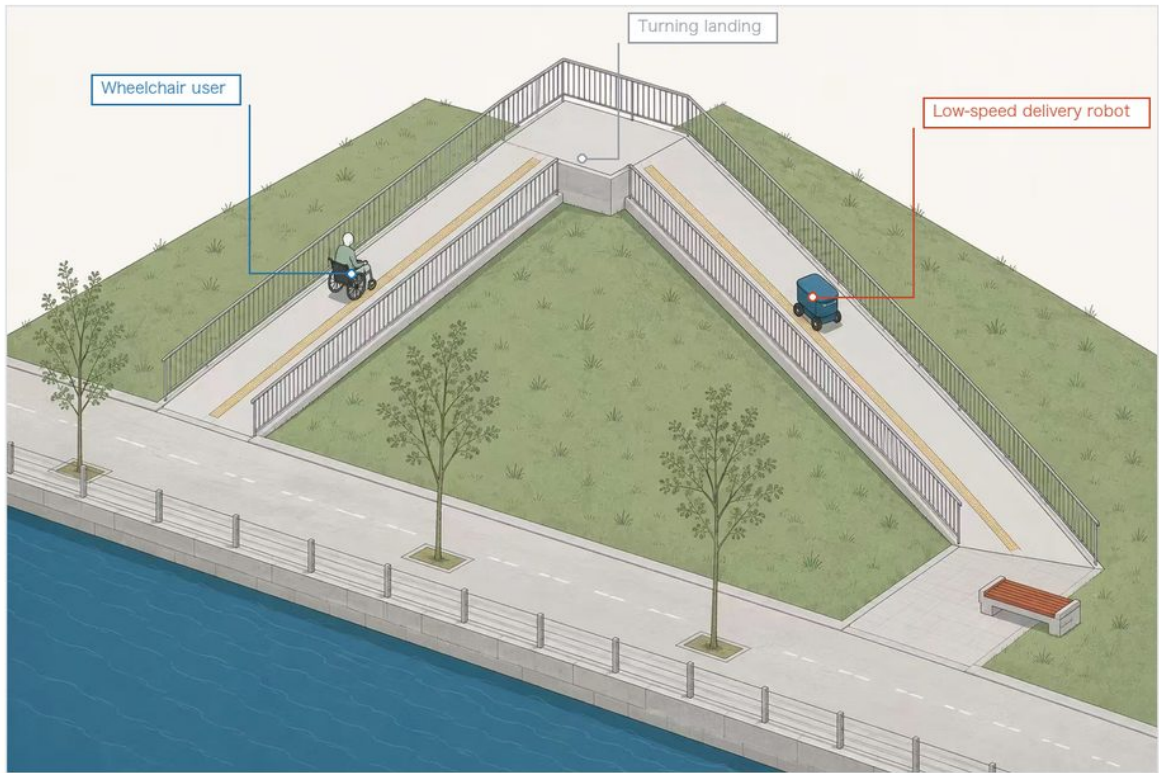
The historical motif is a structural solution, not ornament

The Chevron Ramp — Component Plate

One geometry, a century apart, answering the same constraint

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The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

Where the geometry comes from

Guangou switchback

Accessible ramp in plan



Why the same geometry

Both face the same constraint: gain a fixed height, within a limited depth, at a limited gradient. Switching back is the only answer to it. Guangou used it in 1905 to get a train up the grade; here it gets a wheelchair and a low-speed robot up the same kind of grade. The historical motif is a structural solution, not applied decoration.

What this plate does not give

No gradient, clear width, turning radius or construction detail is given here. Those must be developed by a professional team against current accessibility design codes; this proposal defines only the component type and its function. Putting numbers on the plate would contradict the text.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

Why the same geometry

The Guangou switchback and an accessible ramp face one constraint: climbing a fixed rise at a limited gradient within a limited depth. Switching back is the only answer, which is why they take the same form.

Who it serves at once

Wheelchair users and low-speed delivery robots are bound by the same gradient limit, so one ramp answers both. If the robots leave, the ramp still serves wheelchairs, prams and luggage. This is where the reversibility principle lands.

What this sheet does not give

Gradient, clear width, turning radius and structure are not specified here; they belong to a professional team. The sheet carries only figures this proposal measured and rules its text already states. The illustration is an AI-generated concept image, not a photograph.

Component 2: Platform Unit

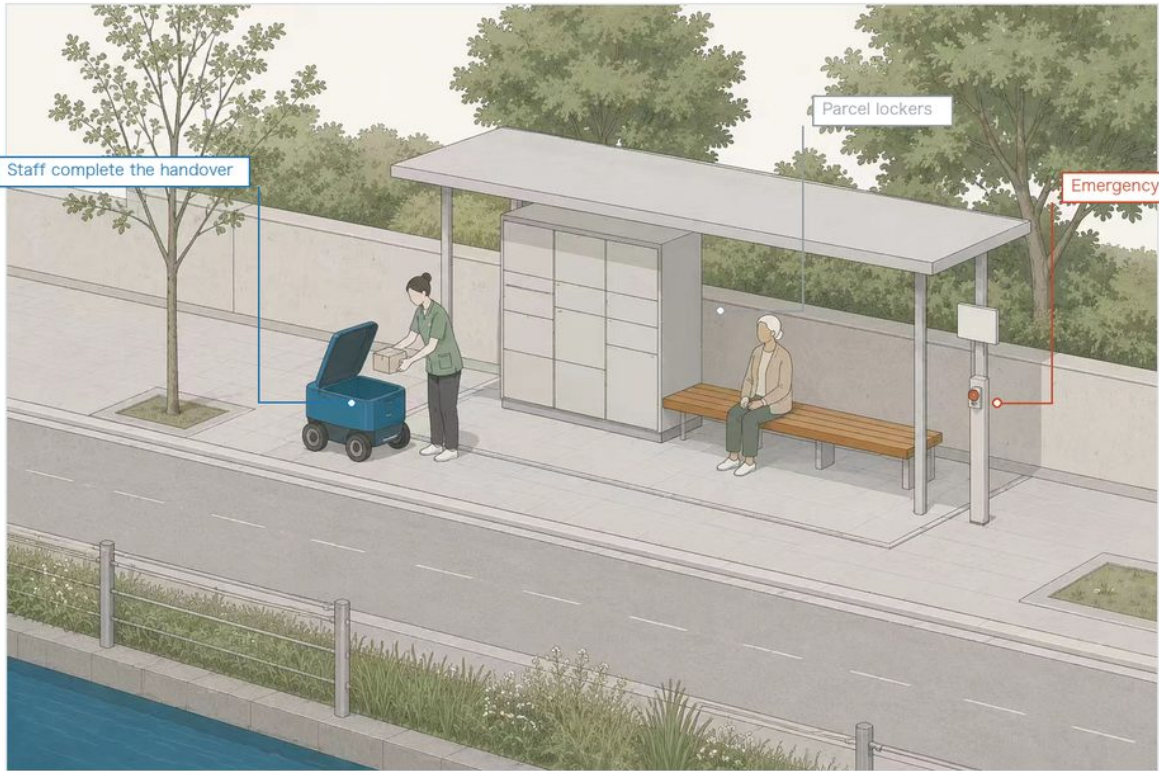
The phase-one deliverable; the handover boundary drives the form

The Platform Unit — Phase One Plate

Phase one is not all five connected: it is 82 m, one stop, one facility

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The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

This stop is a computed result

Phase-one link
82 m
Facilities served
1 / 5
Worst run once merged
14.0 min
Terminal access, five sites
1,950 m

The handover boundary is built into the component

The robot stops at the door and staff take delivery; it does not enter the building, does not interact with the resident directly, and makes no automated decision. Those three rules are not a line in an operations manual — they are why the unit is shaped this way, opening only to the street side with no opening on the building side.

Why phase one serves only one

Terminal distances for the five facilities are 109, 218, 420, 551 and 652 m. Drawing the line at 150 m, the distance at which a round trip to collect a delivery does not become a burden, only one qualifies. The remaining 1,841 m across four sites is walking and accessibility work, separately scoped and scheduled. Writing phase one as "all five connected" would look better and could not be built.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

Phase one serves one facility, and that is computed

Phase one: 82 m of link, one facility of five, 14.0 minutes to the furthest once joined. The other four have 1,841 m of terminal gap between them, past the 150 m threshold, so they belong to walking-and-cycling works rather than equipment procurement.

The boundary is not a note, it is the form driver

No entry indoors, no contact with the resident, no automated decisions. The cabinet therefore opens only to the street and has no indoor-side aperture: staff take receipt on the street side, and the robot has no physical means of entering.

Reuse, not new build

All six robot scenario cards use existing facilities; none requires a dedicated new structure. The platform unit is a demountable module: when the pilot ends it comes out, leaving no permanent occupation of the footway.

Component 3: Observation Point

Where explainability stops being a principle and becomes a requirement

The Observation Point — Explainability Plate

Open to watch is not open data: oversight starts with being able to see what is happening

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The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

Three things the panel must answer

1. What it is doing
2. How the data is handled
3. Who supervises it

Where the three signage principles land

Signage for AI installations cannot simply state a name. The explainability-first principle becomes a concrete requirement here: the panel must answer the three questions above. The observation point itself collects no visitor data — open to watch is not open data. Entry is ramped with no steps, on the same accessibility baseline as the chevron ramp.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

The sign must answer three questions

What is it doing; how is data handled; who supervises it. A device missing any of the three does not enter a public-space pilot. It is the only hard requirement in the wayfinding system.

What is not collected

No faces, no individual movement traces. An observation point is where people watch robots, not where robots watch people: that directional distinction sets its orientation, seating and sightlines.

Why it needs a place of its own

Public acceptance is a risk shared by all ten scenario cards. Putting the robots somewhere people can stop, ask and object works better than explaining in a brochure. The illustration is an AI-generated concept image, not a photograph.

Renewal Project List and Phasing

JZ-07 is the smallest and earliest item

ID	Project	Phase	Type
JZ-07	Field verification of three key network gaps	Near term	Mobility
JZ-01	Heritage-park slow-mobility stitching	Near term	Public space
JZ-02	Zhongzhiyuan Qinghe innovation interface	Medium term	Blue-green
JZ-03	Campus-adjacent translation street	Medium term	Renewal
JZ-04	Dazhongsi four-quadrant pedestrian links	Medium term	Transit
JZ-05	AI service and edge-compute nodes	Long term	Infrastructure
JZ-06	Global AI week public route	Ongoing	Operations

Why JZ-07 comes first

Measurement shows that 56 m of connection across three points could grow the usable robot network from 27,560 m to 57,770 m, a return no other item on the list approaches. It must nevertheless begin with field verification: whether the breaks are mapping artefacts, whether railings or grade changes intervene, and whether kerb gradient and clear width permit passage can only be judged on site. It proceeds only if verification confirms the breaks, and is struck from the list otherwise. Almost all of its cost is verification, not construction.